

A Analytical Structure of Nonlinear IPO Extensions

A.1 Why this appendix matters

This appendix clarifies the analytical structure behind the nonlinear and regularized IPO extensions used in the empirical study. The main distinction is not whether a predictor is nonlinear in the input features, but whether it is linear in its trainable parameters. Polynomial and finite-kernel predictors are nonlinear in the input features but linear in their trainable coefficients, so their IPO objectives retain a quadratic structure. Neural networks, by contrast, are nonlinear in their parameters and therefore introduce nonconvex optimization. This distinction helps interpret the empirical results: poor performance of a nonlinear model may come from optimization instability and high-variance predictions rather than from nonlinearity itself.

A.2 Closed-form MVO decision layer

We consider the integrated prediction-and-optimization problem

$$\min_{\theta} \frac{1}{T} \sum_{t=1}^T c(z_t^*(\hat{y}_t), y_t), \quad \hat{y}_t = f(X_t; \theta),$$

where the downstream mean-variance portfolio decision is

$$z_t^*(\hat{y}_t) = \arg \min_{z \in \mathcal{S}} \left\{ -z^\top \hat{y}_t + \frac{\delta}{2} z^\top V_t z \right\}.$$

In the equality-constrained case

$$\mathcal{S} = \{z : \mathbf{1}^\top z = 1\},$$

the optimal portfolio has the closed form

$$z_t^*(\hat{y}_t) = \frac{1}{\delta} V_t^{-1} (\hat{y}_t - \lambda_t \mathbf{1}),$$

where

$$\lambda_t = \frac{\mathbf{1}^\top V_t^{-1} \hat{y}_t - \delta}{\mathbf{1}^\top V_t^{-1} \mathbf{1}}.$$

Equivalently, the MVO solution is affine in the predicted return vector:

$$z_t^*(\hat{y}_t) = B_t \hat{y}_t + a_t,$$

where

$$B_t = \frac{1}{\delta} \left[V_t^{-1} - \frac{V_t^{-1} \mathbf{1} \mathbf{1}^\top V_t^{-1}}{\mathbf{1}^\top V_t^{-1} \mathbf{1}} \right],$$

and

$$a_t = \frac{V_t^{-1} \mathbf{1}}{\mathbf{1}^\top V_t^{-1} \mathbf{1}}.$$

The key implication is that the MVO optimizer $z_t^*(\hat{y}_t)$ is closed-form for all predictors because it depends only on the predicted return vector $\hat{y}_t$. However, the predictor parameter estimator is closed-form only when $f(X_t; \theta)$ is linear in the trainable parameters and the penalty is smooth.

A.3 Parameter-linear and regularized IPO models

Whenever the predictor is linear in its trainable parameters, the IPO objective remains quadratic in those parameters after substituting the affine MVO decision rule. This section derives the resulting structure for Ridge, Lasso, Elastic Net, polynomial features, and finite-kernel predictors.

A.3.1 Ridge IPO

Consider a linear predictor

$$\hat{y}_t = X_t \beta,$$

where $X_t \in \mathbb{R}^{n_t \times d}$, $\beta \in \mathbb{R}^d$, and $\hat{y}_t \in \mathbb{R}^{n_t}$. Since the MVO solution is affine in $\hat{y}_t$, we have

$$z_t^*(\hat{y}_t) = B_t X_t \beta + a_t.$$

The realized MVO cost is

$$c(z_t^*, y_t) = -(z_t^*)^\top y_t + \frac{\delta}{2} (z_t^*)^\top V_t z_t^*.$$

Substituting $z_t^* = B_t X_t \beta + a_t$, we obtain

$$c_t(\beta) = -(B_t X_t \beta + a_t)^\top y_t + \frac{\delta}{2} (B_t X_t \beta + a_t)^\top V_t (B_t X_t \beta + a_t).$$

Ignoring terms independent of β , the objective can be written as

$$\mathcal{L}(\beta) = \frac{1}{2} \beta^\top H \beta - g^\top \beta + \frac{\lambda}{2} \|\beta\|_2^2,$$

where

$$H = \delta \sum_{t=1}^T X_t^\top B_t^\top V_t B_t X_t,$$

and

$$g = \sum_{t=1}^T X_t^\top B_t^\top (y_t - \delta V_t a_t).$$

The ridge-regularized IPO estimator satisfies the first-order condition

$$(H + \lambda I) \beta = g.$$

Therefore, the closed-form ridge IPO solution is

$$\boxed{\hat{\beta}_{\text{ridge IPO}} = (H + \lambda I)^{-1} g.}$$

A.3.2 Lasso IPO

The Lasso IPO objective adds an L_1 penalty to the IPO loss:

$$\mathcal{L}(\beta) = \frac{1}{2}\beta^\top H\beta - g^\top \beta + \lambda\|\beta\|_1.$$

Equivalently,

$$\hat{\beta}_{\text{lasso IPO}} = \arg \min_{\beta} \left\{ \frac{1}{2}\beta^\top H\beta - g^\top \beta + \lambda\|\beta\|_1 \right\}.$$

Because the L_1 norm is not differentiable at zero, the estimator generally does not admit a simple closed-form expression like ridge regression. The optimality condition is the subgradient condition

$$0 \in H\hat{\beta} - g + \lambda\partial\|\hat{\beta}\|_1.$$

Componentwise, this means

$$(H\hat{\beta} - g)_j + \lambda s_j = 0,$$

where

$$s_j = \begin{cases} \text{sign}(\hat{\beta}_j), & \hat{\beta}_j \neq 0, \\ [-1, 1], & \hat{\beta}_j = 0. \end{cases}$$

Thus, the Lasso IPO estimator is characterized by the KKT conditions:

$$\begin{cases} (H\hat{\beta} - g)_j = -\lambda \text{sign}(\hat{\beta}_j), & \hat{\beta}_j \neq 0, \\ |(H\hat{\beta} - g)_j| \leq \lambda, & \hat{\beta}_j = 0. \end{cases}$$

Only in the special case where H is diagonal does the solution reduce to coordinatewise soft-thresholding:

$$\hat{\beta}_j = \frac{1}{H_{jj}} S(g_j, \lambda),$$

where

$$S(g_j, \lambda) = \text{sign}(g_j) \max\{|g_j| - \lambda, 0\}.$$

In the general IPO setting, H is not diagonal, so Lasso IPO is solved by coordinate descent or proximal gradient methods rather than by a closed-form matrix inverse.

A.3.3 Elastic Net IPO

The Elastic Net IPO objective is

$$\mathcal{L}(\beta) = \frac{1}{2}\beta^\top H\beta - g^\top \beta + \lambda_1\|\beta\|_1 + \frac{\lambda_2}{2}\|\beta\|_2^2.$$

Combining the quadratic terms gives

$$\mathcal{L}(\beta) = \frac{1}{2}\beta^\top (H + \lambda_2 I)\beta - g^\top \beta + \lambda_1 \|\beta\|_1.$$

Therefore,

$$\hat{\beta}_{\text{EN IPO}} = \arg \min_{\beta} \left\{ \frac{1}{2}\beta^\top (H + \lambda_2 I)\beta - g^\top \beta + \lambda_1 \|\beta\|_1 \right\}.$$

The KKT condition is

$$0 \in (H + \lambda_2 I)\hat{\beta} - g + \lambda_1 \partial \|\hat{\beta}\|_1.$$

Componentwise,

$$\boxed{\begin{cases} ((H + \lambda_2 I)\hat{\beta} - g)_j = -\lambda_1 \text{sign}(\hat{\beta}_j), & \hat{\beta}_j \neq 0, \\ |((H + \lambda_2 I)\hat{\beta} - g)_j| \leq \lambda_1, & \hat{\beta}_j = 0. \end{cases}}$$

As with Lasso IPO, the Elastic Net IPO estimator generally does not have a closed-form matrix-inverse solution because of the non-differentiability of the L_1 term. It is typically solved by coordinate descent, proximal gradient, or subgradient-based methods. When $\lambda_1 = 0$, Elastic Net reduces to Ridge IPO and admits the closed form

$$\hat{\beta} = (H + \lambda_2 I)^{-1}g.$$

A.3.4 Polynomial IPO

A polynomial predictor uses a nonlinear feature map

$$\phi(x) = [x, x^{\circ 2}, \{x_j x_k\}_{j < k}],$$

where $x^{\circ 2}$ denotes elementwise squares and $\{x_j x_k\}_{j < k}$ denotes pairwise interaction terms.

For a cross-section at time t , let

$$\Phi_t = \phi(X_t)$$

denote the transformed feature matrix. The polynomial predictor is

$$\hat{y}_t = \Phi_t \beta.$$

Although the predictor is nonlinear in the original input features X_t , it is linear in the parameter vector β . Therefore, using

$$z_t^*(\hat{y}_t) = B_t \hat{y}_t + a_t,$$

we obtain

$$z_t^* = B_t \Phi_t \beta + a_t.$$

The realized MVO cost becomes

$$c_t(\beta) = -(B_t \Phi_t \beta + a_t)^\top y_t + \frac{\delta}{2} (B_t \Phi_t \beta + a_t)^\top V_t (B_t \Phi_t \beta + a_t).$$

Collecting the quadratic and linear terms across t , define

$$H_\phi = \delta \sum_{t=1}^T \Phi_t^\top B_t^\top V_t B_t \Phi_t,$$

and

$$g_\phi = \sum_{t=1}^T \Phi_t^\top B_t^\top (y_t - \delta V_t a_t).$$

Then the polynomial IPO objective can be written as

$$\mathcal{L}(\beta) = \frac{1}{2} \beta^\top H_\phi \beta - g_\phi^\top \beta.$$

The first-order condition is

$$H_\phi \beta = g_\phi.$$

Therefore, the closed-form polynomial IPO solution is

$$\boxed{\hat{\beta}_{\text{poly IPO}} = H_\phi^{-1} g_\phi.}$$

If an L_2 penalty is added, the ridge-polynomial IPO solution becomes

$$\boxed{\hat{\beta}_{\text{poly ridge IPO}} = (H_\phi + \lambda I)^{-1} g_\phi.}$$

A.3.5 Finite-kernel IPO

Consider a finite-dimensional kernel predictor with fixed anchor points $\{u_m\}_{m=1}^M$. Define the kernel feature matrix

$$K_t = [K(x_{i,t}, u_m)]_{i=1, \dots, n_t; m=1, \dots, M}.$$

The kernel predictor is

$$\hat{y}_t = K_t \alpha + b \mathbf{1}.$$

After absorbing the intercept into an augmented kernel matrix

$$\tilde{K}_t = \begin{bmatrix} K_t & \mathbf{1} \end{bmatrix}, \quad \tilde{\alpha} = \begin{bmatrix} \alpha \\ b \end{bmatrix},$$

we can write

$$\hat{y}_t = \tilde{K}_t \tilde{\alpha}.$$

Since the predictor is linear in the parameter vector $\tilde{\alpha}$, the MVO solution becomes

$$z_t^* = B_t \tilde{K}_t \tilde{\alpha} + a_t.$$

The realized IPO objective is therefore quadratic in $\tilde{\alpha}$:

$$\mathcal{L}(\tilde{\alpha}) = \frac{1}{2} \tilde{\alpha}^\top H_K \tilde{\alpha} - g_K^\top \tilde{\alpha},$$

where

$$H_K = \delta \sum_{t=1}^T \tilde{K}_t^\top B_t^\top V_t B_t \tilde{K}_t,$$

and

$$g_K = \sum_{t=1}^T \tilde{K}_t^\top B_t^\top (y_t - \delta V_t a_t).$$

With kernel ridge regularization on the non-intercept parameters, let

$$D = \begin{bmatrix} I_M & 0 \\ 0 & 0 \end{bmatrix}.$$

The regularized kernel IPO objective is

$$\mathcal{L}_{\text{KRR}}(\tilde{\alpha}) = \frac{1}{2} \tilde{\alpha}^\top H_K \tilde{\alpha} - g_K^\top \tilde{\alpha} + \frac{\lambda}{2} \tilde{\alpha}^\top D \tilde{\alpha}.$$

The first-order condition is

$$(H_K + \lambda D) \tilde{\alpha} = g_K.$$

Thus, the finite-dimensional kernel ridge IPO estimator has the closed form

$$\boxed{\hat{\tilde{\alpha}}_{\text{kernel IPO}} = (H_K + \lambda D)^{-1} g_K.}$$

This closed form relies on the anchor points being fixed. If the kernel parameters, such as the RBF bandwidth γ , are also trained, then the problem is no longer linear in the parameters and generally requires gradient-based optimization.

A.4 Fully nonlinear IPO models

A.4.1 Neural Network IPO

A single-hidden-layer neural network predictor takes the form

$$\hat{y}_t = f(X_t; \theta) = W_2 \sigma(W_1 X_t + b_1) + b_2,$$

where

$$\theta = \{W_1, b_1, W_2, b_2\}.$$

The MVO solution is still affine in the predicted return vector:

$$z_t^* = B_t f(X_t; \theta) + a_t.$$

Therefore, the IPO objective is

$$\mathcal{L}(\theta) = \frac{1}{T} \sum_{t=1}^T \left[- (B_t f(X_t; \theta) + a_t)^\top y_t + \frac{\delta}{2} (B_t f(X_t; \theta) + a_t)^\top V_t (B_t f(X_t; \theta) + a_t) \right].$$

Unlike the polynomial or finite-kernel cases, $f(X_t; \theta)$ is nonlinear in θ . As a result, $\mathcal{L}(\theta)$ is generally nonconvex and does not admit a closed-form estimator.

The neural-network IPO estimator is therefore defined as

$$\hat{\theta}_{\text{NN IPO}} = \arg \min_{\theta} \mathcal{L}(\theta)$$

and is solved by gradient-based optimization.

The gradient is computed through the differentiable MVO layer:

$$\nabla_{\theta} \mathcal{L}(\theta) = \sum_{t=1}^T \left(\frac{\partial c_t}{\partial z_t^*} \frac{\partial z_t^*}{\partial \hat{y}_t} \frac{\partial f(X_t; \theta)}{\partial \theta} \right).$$

Since

$$z_t^*(\hat{y}_t) = B_t \hat{y}_t + a_t,$$

we have

$$\frac{\partial z_t^*}{\partial \hat{y}_t} = B_t.$$

Therefore,

$$\nabla_{\theta} \mathcal{L}(\theta) = \sum_{t=1}^T \left[(-y_t + \delta V_t z_t^*)^\top B_t \frac{\partial f(X_t; \theta)}{\partial \theta} \right],$$

which can be evaluated by automatic differentiation. This is the approach used for neural-network IPO.

A.5 Summary of analytical structure

Table 1 summarizes which model classes preserve a closed-form or convex IPO structure.

The key point is that the MVO optimizer $z^*(\hat{y}_t)$ is closed-form for all predictors because it depends only on the predicted return vector $\hat{y}_t$. However, the predictor parameter estimator is closed-form only when $f(X_t; \theta)$ is linear in the trainable parameters and the penalty is smooth. This distinction explains why ridge, polynomial, and finite-kernel models have closed-form IPO structure, while Lasso and Elastic Net require nonsmooth convex optimization and neural networks require gradient-based nonconvex optimization.

Model	Nonlinear in inputs?	Linear in parameters?	Closed-form IPO estimator?
Linear	No	Yes	Yes
Ridge	No	Yes	Yes
Lasso	No	Yes	No, due to L_1 nonsmoothness
Elastic Net	No	Yes	No, unless $\lambda_1 = 0$
Polynomial	Yes	Yes	Yes
Finite Kernel Ridge	Yes	Yes, if anchors fixed	Yes
Neural Network	Yes	No	No

Table 1: Closed-form availability for IPO extensions. The key distinction is not whether the predictor is nonlinear in the inputs, but whether it is linear in the trainable parameters.

A.6 Connection to implementation and empirical results

The derivations above clarify the analytical structure of the IPO objective. In the implementation, however, we use a unified training interface. The OLS plug-in baselines use closed-form least squares when available and sklearn coordinate descent for Lasso and Elastic Net. The IPO variants are trained end-to-end by backpropagating realized MVO cost through the closed-form equality-constrained MVO layer. Thus, the closed form is used operationally for $z^*(\hat{y}_t)$ in every IPO model, while the predictor parameters are updated by gradient descent except in the OLS plug-in baselines.

The theoretical classification also helps explain the empirical ranking. Polynomial and finite-kernel predictors are flexible but high-dimensional; even though their IPO objectives retain a tractable structure, the lifted feature space can amplify estimation noise in $\hat{y}_t$, which is then magnified by the MVO layer into extreme portfolio weights. In contrast, Lasso and Elastic Net impose sparsity directly on the predictor, reducing the variance of $\hat{y}_t$ and producing more stable portfolio decisions.

Table 2 maps the theoretical structure to the implemented model classes.

Model	Predictor class	Analytical structure	
Linear/Ridge	Parameter-linear	Quadratic IPO objective	OLS
Lasso/Elastic Net	Parameter-linear + L_1	Convex but nonsmooth	sklearn
Polynomial	Nonlinear features, parameter-linear	Quadratic after feature expansion	Feat
Finite Kernel	Fixed-anchor nonlinear features	Quadratic in kernel weights	Fixed an
Neural Network	Nonlinear in parameters	Nonconvex IPO objective	Backp

Table 2: How the theoretical IPO structure maps to the implemented model classes.

A.7 Covariance estimation

At each monthly rebalance date t , the MVO decision requires a covariance matrix V_t whose units are consistent with the realized next-month return vector y_t . Since y_t is measured at the monthly horizon, we estimate a daily covariance matrix from recent daily returns and then rescale it to a monthly covariance matrix.

Let $r_\tau \in \mathbb{R}^{n_t}$ denote the vector of daily returns for the assets in the cross-section at rebalance date t , where n_t is the number of available assets at that rebalance. We use a trailing window of $L = 60$ trading days:

$$\mathcal{W}_t = \{t - L + 1, \dots, t\}.$$

Stacking the demeaned daily returns over this window gives

$$R_t = \begin{bmatrix} (r_{t-L+1} - \bar{r}_t)^\top \\ (r_{t-L+2} - \bar{r}_t)^\top \\ \vdots \\ (r_t - \bar{r}_t)^\top \end{bmatrix} \in \mathbb{R}^{L \times n_t},$$

where

$$\bar{r}_t = \frac{1}{L} \sum_{\tau \in \mathcal{W}_t} r_\tau.$$

The raw sample covariance estimator is

$$S_t = \frac{1}{L} R_t^\top R_t.$$

However, in our setting n_t is large relative to the rolling window length L . For example, the empirical implementation uses approximately $n_t = 100$ assets and only $L = 60$ daily observations. The sample covariance matrix S_t can therefore be ill-conditioned or unstable, which is especially problematic because the MVO solution requires solving linear systems involving V_t .

To stabilize the covariance estimate, we use a Ledoit–Wolf style linear shrinkage estimator toward a scaled identity matrix. Define the shrinkage target

$$F_t = \mu_t I_{n_t}, \quad \mu_t = \frac{1}{n_t} \text{tr}(S_t).$$

Here, F_t preserves the average variance level of the sample covariance but removes noisy off-diagonal correlation estimates. The shrinkage covariance is then

$$\hat{\Sigma}_t^{\text{daily}} = \rho_t F_t + (1 - \rho_t) S_t, \quad \rho_t \in [0, 1].$$

When $\rho_t = 0$, the estimator reduces to the sample covariance S_t . When $\rho_t = 1$, the estimator becomes the scaled identity target F_t . Intermediate values trade off sample correlation information against numerical stability.

In the implementation, the shrinkage intensity is computed from the data. Let

$$d_t^2 = \|S_t - F_t\|_F^2$$

measure the squared Frobenius distance between the sample covariance and the target. The code estimates the total variance of the sample covariance entries by

$$\pi_t = \sum_{i=1}^{n_t} \sum_{j=1}^{n_t} \left[\frac{1}{L} \sum_{\tau \in \mathcal{W}_t} \tilde{r}_{\tau,i}^2 \tilde{r}_{\tau,j}^2 - (S_t)_{ij}^2 \right],$$

where

$$\tilde{r}_{\tau} = r_{\tau} - \bar{r}_t.$$

The shrinkage intensity is then clipped to the unit interval:

$$\rho_t = \min \left\{ 1, \max \left\{ 0, \frac{\pi_t}{L d_t^2} \right\} \right\}.$$

This produces a well-conditioned covariance estimate even when the rolling window is short relative to the number of assets.

Finally, because the covariance estimate above is based on daily returns while the realized return vector y_t is a next-month return, we rescale the daily covariance to a monthly horizon:

$$V_t = 21 \hat{\Sigma}_t^{\text{daily}}.$$

The factor 21 approximates the number of trading days in one month. This rescaling ensures that the two terms in the MVO objective,

$$-z^\top \hat{y}_t \quad \text{and} \quad \frac{\delta}{2} z^\top V_t z,$$

are expressed in compatible monthly units. Thus, at each rebalance date, the portfolio weights are computed using the estimated monthly covariance matrix V_t , the predicted next-month return vector $\hat{y}_t$, and the fixed risk-aversion parameter δ .

A.8 Limitations of the analytical structure

The derivations characterize the optimization structure conditional on a fixed covariance estimate V_t . They do not solve the joint problem of learning both expected returns and the covariance matrix, nor do they incorporate long-only, leverage, or turnover constraints. These extensions would generally destroy the simple affine MVO layer and require either differentiable quadratic programming layers or approximate projection methods.